LAB # 11

Interfacing ADC with 8051

Spring 2024

CSE-307L Microprocessor Based System Design Lab

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Class Section: **C**

"On my honor, as student at University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work."

Submitted to:

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Objectives of The Lab:

ADC with 8051 microcontroller

Lab Total Tasks:

From temperature sensor (LM35) read temperature and convert it into Digital value by using ADC0804 and display the value on the LCD. In LCD at first line write your registration Number and on the second line display the value of the temperature sensor attached with ADC0804.

Solution:

C Code:

```c
#include<reg51.h>
sbit E = P0^7;
sbit RW = P0^6;
sbit RS = P0^5;
sbit READ  = P3^1;
sbit WRITE = P3^2;
sbit INTERRUPT = P3^3;

void writeData(char d);
void writeCommand(char cmd);
double adc();
void initialize_LCD();
void E_Pulse();
void Delay(unsigned int n);
void writeString(char* d);
void disp_temp(double num);
double adc_Value;
double temp;

```

```c
void main(){
		P2=0;
		initialize_LCD();
		writeString("21PWCSE1980");

	while(1){
		adc_Value = adc();
		writeCommand(0xC0);
		writeString("REDING ");
		disp_temp(adc_Value);
		}
}
```

```c
double adc ()
{

    READ=1;
    WRITE=0;
    Delay(1);
    WRITE=1;
    while (INTERRUPT==1);
    Delay(1);
    READ=0;
    temp=P1;
    Delay(3);
    return temp;
}
```

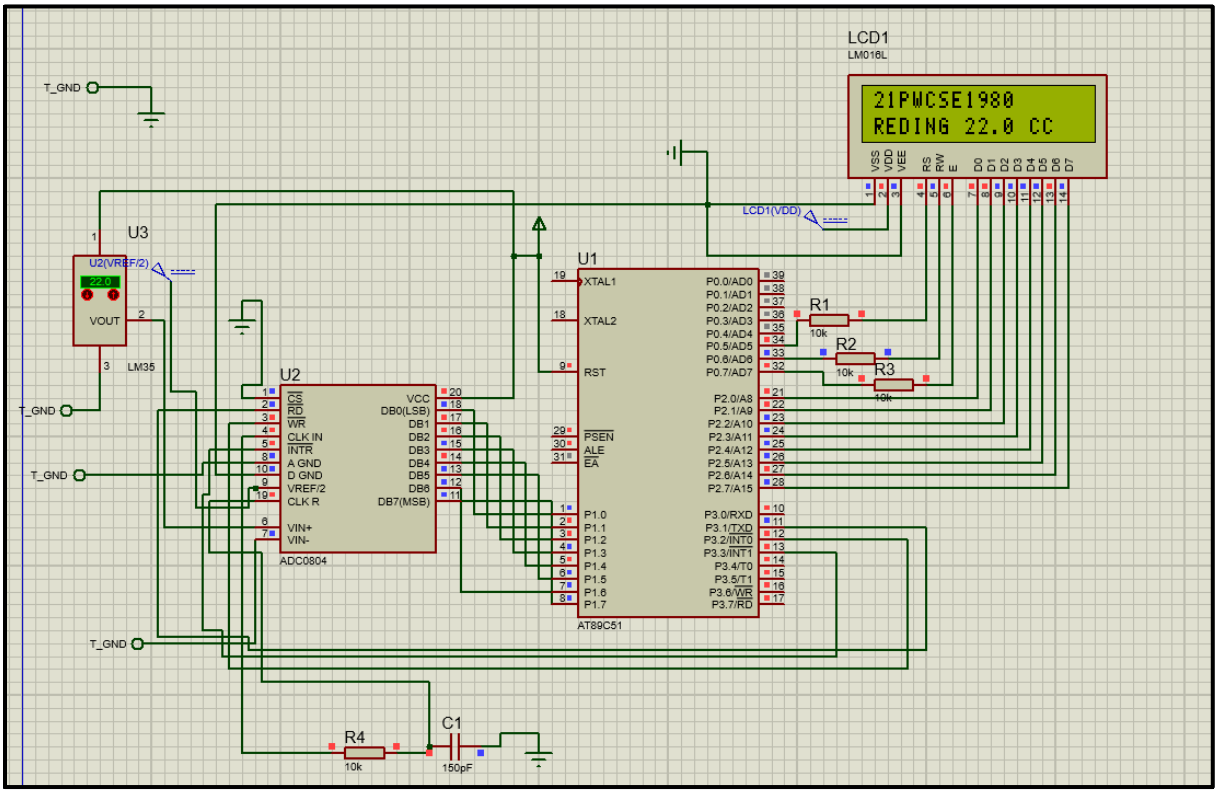
```c
void disp_temp(double num) {
  unsigned char UnitDigit = 0;
  unsigned char TenthDigit = 0;
  unsigned char HundDigit = 0;
  unsigned char decimal = 0;
  unsigned char point;

  point= num*10;
  HundDigit = (num/100);
  if(HundDigit!=0){
      writeData(HundDigit + 0x30);
  }
  TenthDigit = (num - HundDigit*100);
  TenthDigit = TenthDigit/10;

  if(HundDigit ==0 && TenthDigit == 0){}
  else{
      writeData(TenthDigit + 0x30);
  }
  UnitDigit = (num - HundDigit*100);
  UnitDigit = (UnitDigit - TenthDigit*10);
  writeData(UnitDigit+0x30);
```

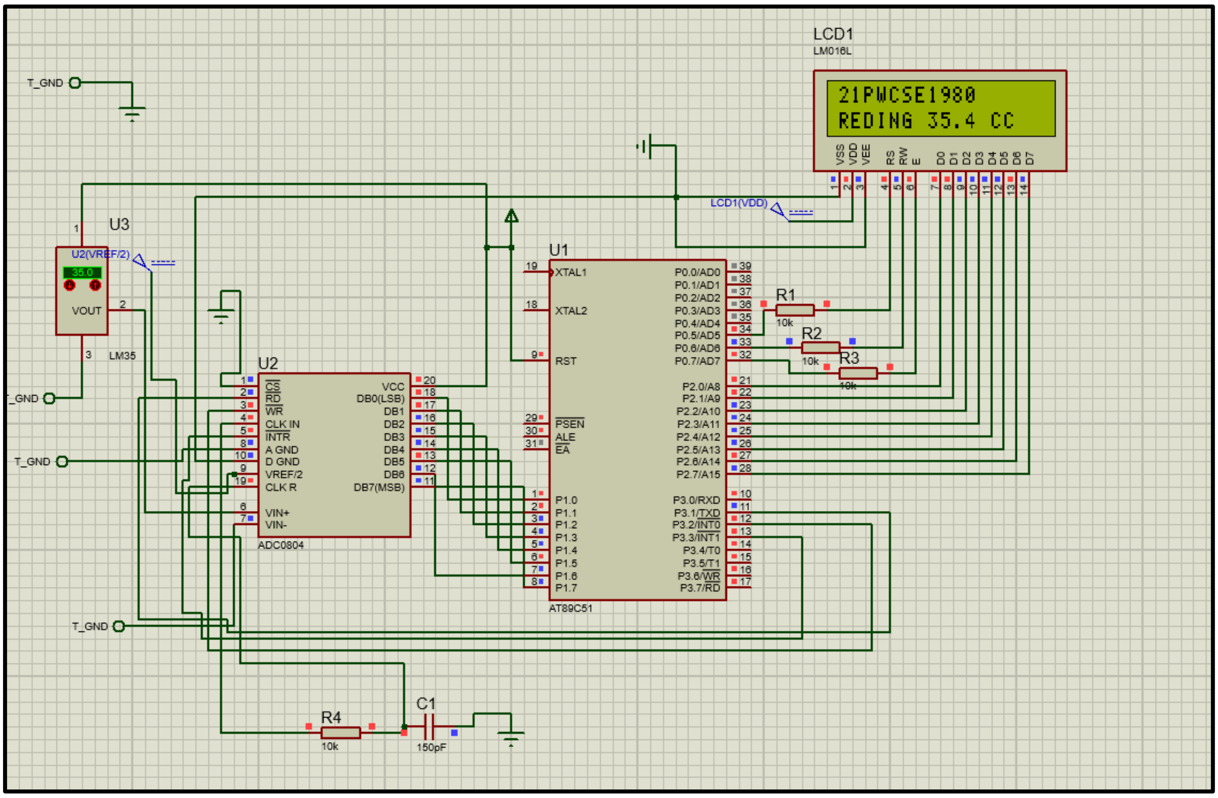
```
    writeData('.');
    decimal = (point%10);
    writeData(decimal + 0x30);
    writeData(' ');
    writeData('C');

}


void initialize_LCD(){
        writeCommand(0x38); //2 lines & 5x7 Matrix
        writeCommand(0x0E); // Display on and cursor blinki
        writeCommand(0x01); // clr dsp;
        writeCommand(0x0C);
        writeCommand(0x80);
}
```

```
void writeData(char d){
        RW=0;
        P2=d;
        RS=1;
        E_Pulse();
}


void writeString(char* d){

        while(*d){
            writeData(*d);
            *d++;
        }
}
```

```c
void writeCommand(char cmd){
      RW=0;
      RS=0;
      P2=cmd;
      E_Pulse();
}

void E_Pulse(){
      E=1;
      Delay(4000);
      E=0;
      Delay(4000);
}

void Delay(unsigned int n){
      unsigned int i;
      for(i=0;i<n;i++);
}
```

Proteus Simulation:





LCD1
LM016L
21PWCSE1980
REDING 35.4 CC
U1
AT89C51
U2
ADC0804
U3
LM35
R1
R2
R3
R4
C1
150pF
10k
T_GND
LCD1(VDD)
U2(VREF/2)